

CTRhythm: Accurate Atrial Fibrillation Detection from Single-Lead ECG by Convolutional Neural Network and Transformer Integration

Zhanyu Liang

Department of Statistics and Financial Mathematics, School of Mathematics
South China University of Technology
Guangzhou, China
maliangzy@mail.scut.edu.cn

Chen Yang

Department of Cardiology
Sun Yat-sen Memorial Hospital of Sun Yat-sen University
Guangzhou, China
yangy556@mail.sysu.edu.cn

Zhengyang Yu

Department of Statistics and Financial Mathematics, School of Mathematics
South China University of Technology
Guangzhou, China
202220129909@mail.scut.edu.cn

Yinmingren Fu

Department of Statistics and Financial Mathematics, School of Mathematics
South China University of Technology
Guangzhou, China
mafuymr@mail.scut.edu.cn

Bozhen Ren

Department of Statistics and Financial Mathematics, School of Mathematics
South China University of Technology
Guangzhou, China
mabren@mail.scut.edu.cn

Maohuan Lin

Department of Cardiology
Sun Yat-sen Memorial Hospital of Sun Yat-sen University
Guangzhou, China
lmhuan@mail.sysu.edu.cn

Qingjiao Li

The Eighth Affiliated Hospital
Sun Yat-sen University
Shenzhen, China
liqj23@mail.sysu.edu.cn

Xuemei Liu

School of Physics and Optoelectronics
South China University of Technology
Guangzhou, China
liuxm@scut.edu.cn

Yangxin Chen

Department of Cardiology
Sun Yat-sen Memorial Hospital of Sun Yat-sen University
Guangzhou, China
chenyx39@mail.sysu.edu.cn

Li C. Xia

Department of Statistics and Financial Mathematics, School of Mathematics
South China University of Technology
Guangzhou, China
lcxia@scut.edu.cn

Abstract—Atrial Fibrillation (AF) is a common supraventricular arrhythmia that affects about 30 million people globally. Electrocardiogram (ECG) analysis is the primary diagnostic approach. The widespread adoption of wearable devices for heart rhythm monitoring prompted the development of AF detection models for single-lead ECGs, benefitting real-time early diagnosis. Current state-of-the-art methods for AF detection are convolutional neural network (CNN) and convolutional recurrent neural network (CRNN) based models, which only focus on capturing local patterns despite heart rhythms exhibiting rich long-range dependencies. To address this limitation, we propose a novel method for single-lead ECG rhythm classification, termed CNN-Transformer Rhythm Classifier (CTRhythm), which integrates CNN with a Transformer encoder to effectively capture both local and global patterns effectively. CTRhythm achieved an overall F1 score of 0.831, outperforming baseline deep learning models on the gold-standard CINC2017 dataset. Furthermore, pre-training with additional data improved the overall F1 score to 0.840. In two external validation datasets, CTRhythm showed its strong generalization capabilities. CTRhythm is freely available at <https://github.com/labxscut/CTRhythm>.

Keywords—Atrial fibrillation, ECG classification, Convolutional neural network, Transformer encoder

I. INTRODUCTION

Atrial Fibrillation (AF) is a supraventricular tachyarrhythmia that affects approximately 30 million people globally [1]. Timely detection of AF is essential for effectively managing heart conditions and preventing severe consequences [2]. Electrocardiogram (ECG) has long been an

indispensable diagnostic tool in identifying cardiac abnormalities, including AF. In recent years, the rapid emergence of personal cardiac monitoring devices, such as smartwatches, has led to a significant increase in single-lead ECG data [3]. Compared to 12-lead ECG data collected in hospitals, single-lead ECG data is more readily accessible through portable wearable devices. Once anomalies are detected on wearable devices, users can be prompted to seek more detailed examinations at the hospital, facilitating timely early diagnosis and treatment.

In machine learning, AF detection is commonly framed as a heart rhythm classification task [4-6]. Recently, the advent of deep neural networks significantly improved classification performance by enabling automatic feature extraction and end-to-end classification. Among deep learning methods, the convolutional neural networks (CNN) [7-13] gained prominence in AF detection, primarily due to their efficient automatic continuous feature extraction capabilities using the convolution operator. However, CNN is inherently designed for extracting local features due to its localized receptive field of convolution kernels, making it less adept at capturing crucial contextual and global dependencies [14]. Researchers tried integrating CNN with recurrent neural networks (RNN) to address this shortcoming and proposed several CRNN-based methods [15-19]. However, those approaches require sequential traversal of the entire ECG sequence, which is inefficient and ineffective when dealing with long sequences [14].

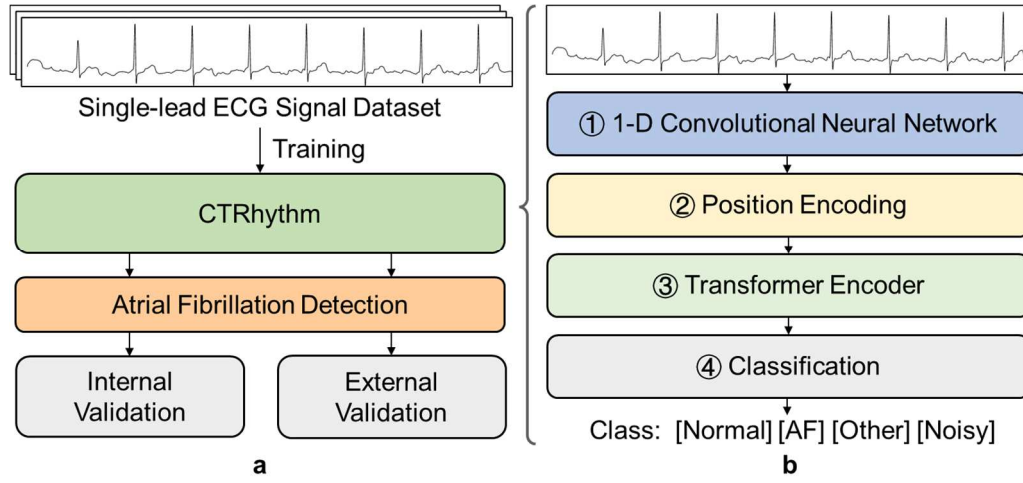


Fig. 1. Overview of the study: a) the training and validation workflow; b) the network architecture overview of CTRhythm.

Transformers have recently emerged as a robust model for effectively learning long-range dependencies [14]. Numerous Transformer-based models achieved remarkable successes in natural language processing [20] and computer vision [21] tasks. They were also applied to ECG data for various tasks [22–24], including atrial fibrillation and flutter segmentation, heartbeat location and detection, and AF classification. Here, we propose adopting and combining Transformers with previously successful CNN models for AF detection, with a focus on performance with single-lead ECG data.

In this work, we integrated CNN and Transformer to create a comprehensive encoder and classification framework for heart rhythm classification, termed CTRhythm (Fig. 1). CTRhythm first employed a CNN to extract local features and then utilizes a Transformer to encode global correspondence. The integration empowers CTRhythm to leverage the strengths of both encoders in capturing short-range and long-range rhythmic relationships. Our contributions are as follows:

- We developed CTRhythm, a new rhythm classifier that effectively integrates Transformer and CNN models for AF detection.
- CTRhythm encodes local features and long-range dependencies, showing higher rhythm classification accuracy over baseline methods in cross-validation.
- Independent validation results with two external datasets showed that CTRhythm can generalize well.
- Ablation experiment results assessing the effect of pre-training indicated that incorporating external data can further improve CTRhythm's performance.

II. RELATED WORKS

A. ECG Classification Methods

AF detection from ECG signals is commonly framed as a classification task [4–6]. In recent years, deep learning models, such as Convolutional Neural Networks and Recurrent Neural Networks, demonstrated significantly improved performance in ECG classification tasks while generally eliminating the need for manual feature extraction.

CNN is the most used deep neural network in ECG classification. For example, Hannun et al. [8] developed a deep CNN to classify 12 rhythm classes in single-lead ECG. Xiong et al. [9] proposed a 16-layer 1D CNN, demonstrating

improved performance by incorporating skip connections for rhythm classification.

CNNs are proven to extract local features efficiently but limited in modeling global contextual information because of their localized receptive field of convolution kernels. To address this limitation, researchers grafted Recurrent Neural Network (RNN) layers, such as Long Short-Term Memory (LSTM) [25], into CNNs. The resulting hybrid model, Convolutional Recurrent Neural Networks (CRNNs), proved moderately effective in classifying heart rhythm. For example, Zihlmann et al. [19] computed a one-sided spectrogram of ECG signals, applied a logarithmic transform, and then combined CNN and LSTM to detect AF.

Although these hybrid models show improvement, the sequential and forgetful nature of RNN layers [14] has deteriorated performance processing long sequences, thus limiting the effective learning of global patterns. Additionally, the requirement for laborious, step-by-step traversal of the entire sequences also hinders RNNs from efficiently encoding long-range dependencies, significantly as the interaction of distant time points diminishes with elongated training steps.

B. Transformer

Transformer represents a modern approach to learning positional correspondence. It leverages a one-step attention mechanism to capture long-range dependencies and process the sequence positions in parallel. Vaswani et al. introduced the first Transformer encoder [14]. Transformers have since undergone extensive testing in natural language processing and computer vision. The proven versatility and capability of Transformers have prompted researchers to explore their potential applications in ECG data analysis. For example, Yun et al. [22] developed a Transformer model for atrial fibrillation and flutter segmentation in long ECGs. Yang et al. [23] decomposed long ECGs into clinical meaningful components and tokenized them. They then used a Transformer model to process these tokens for AF classification. Hu et al. [24] employed a Transformer model to locate and detect heartbeats from ECG signals simultaneously. Nonetheless, these studies have focused on applying Transformers to 12-lead hospital ECG data, and

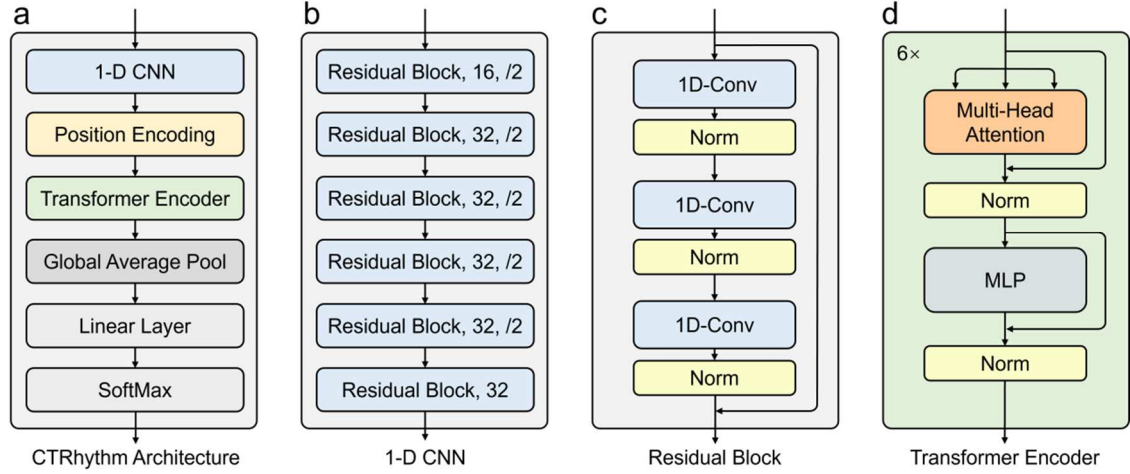


Fig. 1. The CTRhythm model: **a)** its component modules and data flow; **b)** the 1-D CNN module: 16 or 32 is the number of output channels, /2 is the down-sampling rate; **c)** convolution layers within one residual block of 1-D CNN; **d)** the multi-head Transformer encoder module.

their potential value for single-lead data from wearable devices has yet to be exploited.

III. METHODS

A. Overview

We integrated CNN and Transformer to create a novel and effective hybrid AF detection model, CTRhythm. CTRhythm consists of four component modules (see Fig. 2):

1. 1-D Convolutional Neural Network Module (1-D CNN): this module down-samples the input ECG sequence and aggregates the local features using CNN (Fig. 2b).
2. Position Encoding Module: this module applies positional encoding to the CNN-encoded feature sequence to encode the sequence order (Fig. 2a).
3. Transformer Encoder Module: this module performs attention-based learning of long-range dependencies on the position-CNN-encoded sequence (Fig. 2d);
4. Classification Module: this module consists of a global average pooling layer, a linear layer, and a SoftMax layer to predict rhythm class scores and labels (Fig. 2a).

B. Convolutional Neural Network Module

Our CNN module design was inspired by ref. [26]. The critical difference is that we incorporated more down-sampling convolutional layers to reduce the original sequence length more effectively. The CNN module in CTRhythm is comprised of 6 residual blocks. Each basic residual block shall consist of three 1-D convolutional layers. Batch Normalization (BN) [27] and the Rectified Linear Unit (ReLU) activation layers [28] are applied after each convolutional layer (see Fig. 2c).

We employed two types of residual blocks (Fig. 2b). In the first type of residual block, the stride of the convolutional kernel is set to 2, efficiently halving the sequence length. The second type of residual block maintains the shape of the input to the output sequence. Both blocks use convolutional kernels of size 3, and the number of convolutional kernels gradually increases to 32.

We stack six residual blocks to form the CNN module (Fig. 2b). Overall, the ECG sequence transforms into a feature sequence with reduced length and increased dimensions.

C. Position Encoding Module

Unlike RNNs, the Transformer encoder layer neglects the positional order within the sequence [14]. Therefore, preserving the features' order through positional encoding is crucial. Each position in the input feature sequences is linearly projected into an embedding positional latent space of dimension d using a learnable matrix $\mathbf{W}^E \in \mathbb{R}^{k \times d}$. In CTRhythm, d is set to 32, determined through experimental tuning. We used sine and cosine functions [14] as the encoding bases. We denote the positional encoding of the sequence as $\mathbf{E}^{PE}$:

$$\begin{aligned} \mathbf{E}^{PE}(pos, 2i) &= \sin(pos/10000^{2i/d}), \\ \mathbf{E}^{PE}(pos, 2i+1) &= \cos(pos/10000^{2i/d}), \end{aligned} \quad (1)$$

where pos is the position in the sequence, and i is the dimension. $\mathbf{E}^{PE}(pos, 2i)$ denotes the element in the pos row and $2i$ column of the matrix $\mathbf{E}^{PE}$. The formula of position encoding is as follows:

$$\text{PE}(\mathbf{X}) = \mathbf{X}\mathbf{W}^E + \mathbf{E}^{PE}, \quad (2)$$

D. Transformer Encoder Module

We stacked 6 Transformer encoder layers to form the Transformer encoder module. We employed the Multi-head Self-Attention (MSA) mechanism in each layer to capture long-range dependencies. The MSA can attend to information from dimensions and positions simultaneously. In the i -th self-attention component, we generated the matrices $\mathbf{Q}_i, \mathbf{K}_i$ and $\mathbf{V}_i$ as the products of $\mathbf{X}$ and weight matrices $\mathbf{W}_i^Q, \mathbf{W}_i^K$ and $\mathbf{W}_i^V$, then we used a linear projection to fuse the outputs from all heads, as follows:

$$[\mathbf{Q}_i, \mathbf{K}_i, \mathbf{V}_i] = \mathbf{X}[\mathbf{W}_i^Q, \mathbf{W}_i^K, \mathbf{W}_i^V], \quad [\mathbf{W}_i^Q, \mathbf{W}_i^K, \mathbf{W}_i^V] \in \mathbb{R}^{d \times 3d_k},$$

$$\text{SA}_i(\mathbf{X}) = \text{softmax}\left(\frac{\mathbf{Q}_i \mathbf{K}_i^T}{\sqrt{d_k}}\right) \mathbf{V}_i,$$

$$\text{MSA}(\mathbf{X}) = [\text{SA}_1(\mathbf{X}); \text{SA}_2(\mathbf{X}); \dots; \text{SA}_h(\mathbf{X})] \mathbf{W}^{MSA}, \quad (3)$$

where d_k is the dimension of $\mathbf{Q}_i, \mathbf{K}_i$ and $\mathbf{V}_i$, $\mathbf{W}^{MSA} \in \mathbb{R}^{hd_k \times d}$, $i = 1, 2, \dots, h$, and h is the number of attention heads. In our model h is set to 8, and d_k is set to $d/h = 4$.

We used the Multi-Layer Perceptron (MLP) to facilitate additional non-linear transformations.

$$\text{MLP}(\mathbf{X}) = \max(\mathbf{X}\mathbf{W}_1 + b_1, 0) \mathbf{W}_2 + b_2; \quad (4)$$

where $\mathbf{W}_1 \in \mathbb{R}^{d \times d_m}$, $\mathbf{W}_2 \in \mathbb{R}^{d_m \times d}$, and b_1, b_2 are learnable parameters of two distinct linear layers. In our model, the hidden size d_m is set to 256.

After each MSA or MLP layer, Layer Normalization (LN) [29] is applied to normalize the features of each sample, thereby standardizing all input features as follows:

$$\text{LN}(\mathbf{X}) = \frac{\mathbf{X} - \mathbb{E}[\mathbf{X}]}{\sqrt{\text{Var}[\mathbf{X}]}} \gamma' + \beta'; \quad (5)$$

where γ' and β' are learnable parameters adjusting the normalizing effect by this layer, and $\mathbb{E}[\mathbf{X}]$ and $\sqrt{\text{Var}[\mathbf{X}]}$ are the mean and standard deviation computed over the dimensions (along the d subscripts of $\mathbf{X}$).

We denoted the input sequence as $\mathbf{X}$ and the output of the i -th Transformer encoder layer as $\mathbf{X}_i$, $i = 1, 2, \dots, 6$. Let $\mathbf{X}_0 = \mathbf{X}$, $\mathbf{X}_i$ can be iteratively computed as follows:

$$\begin{aligned} \mathbf{X}'_i &= \text{LN}(\text{MSA}(\mathbf{X}_{i-1}) + \mathbf{X}_{i-1}), i = 1, 2, \dots, 6; \\ \mathbf{X}_i &= \text{LN}(\text{MLP}(\mathbf{X}'_i) + \mathbf{X}'_i), i = 1, 2, \dots, 6; \end{aligned} \quad (6)$$

where $\mathbf{X}'_i$ represents the intermediate output at the i -th layer. The last layer's output is the Transformer encoder's output:

$$\text{TransformerEncoder}(\mathbf{X}) = \mathbf{X}_6. \quad (7)$$

E. Classification Module.

The classification module consists of one Global Average Pooling (GAP) [30], one linear, and one SoftMax layer. The selection of all hyperparameters is based on ref. [14], and they are adjustable based on the actual properties of ECG data.

IV. EXPERIMENT

A. Datasets

The Training and Cross-Validation Dataset. The primary dataset used in this study is the PhysioNet/Computing in Cardiology Challenge 2017 (CINC2017) dataset [31]. This dataset comprises ECG recordings collected with the AliveCor, a wearable ECG device capable of recording single-lead ECG data. The training set consists of 8528 single-lead ECG recordings, encompassing four types of heart rhythms: Normal, AF (Atrial Fibrillation), Other rhythm, and Noisy. Following the experimental setup for baseline models in previous works [10, 19, 32, 33], we cross-validated CTRhythm and compared it to other models using this dataset. The average F1 score from 5-fold Cross-Validation is reported as the performance metric.

The Two External Validation Datasets. To evaluate the generalizability of CTRhythm and other models under distribution shifts, such as those caused by switching data sources, we used two external validation datasets: the China Physiological Signal Challenge 2018 (CPSC2018) [34] and the Chapman datasets [35]. Both datasets were collected by hospitals and are publicly available. We extracted all normal and AF ECG records from the two datasets, and then applied models trained on CINC2017 to those datasets.

The Pre-Training Dataset. To further explore the potential of CTRhythm through pre-training, we used the MIMIC-IV-ECG [36] dataset. MIMIC-IV-ECG contains approximately 800,000 12-lead ECG records. We only used those ECGs for self-supervised pre-training.

B. Data Preprocessing

We down-sampled raw ECGs to 150 Hz to reduce the input size and alleviate computational burden while maintaining

performance. We normalized samples by Z-score to eliminate scale differences and mean shifts. The normalized data were truncated or padded to a uniform length of 60 seconds.

C. Evaluation Metrics

We employed standard evaluation metrics from an earlier challenge competition [31]. Considering the suboptimal quality of the Noisy class ECGs, this paper excludes it from computing the final classification performance, as done by the 2017 PhysioNet/CinC Challenge. Performance is assessed based on the average F1 score for the other three categories: Normal (N), AF (A), and Other (O), denoted as $F1_N$, $F1_A$, and $F1_O$. The overall F1 score is denoted as $F1_{All}$, is calculated as the average of the F1 scores for all three categories. We also used accuracy to evaluate a model's performance. In independent external validation, we used accuracy, $F1_A$, $F1_N$, and $F1_{All}$ to evaluate the model performances, where $F1_O$ is not applicable.

D. Implementation Details

The experiments were conducted using PyTorch 2.1. The computational resources for training included an Intel Core i9-12900K CPU, 128 GB RAM, and an NVIDIA RTX 3090 GPU. We designed two groups of experiments to implement CTRhythm:

Training and Validation. All models were trained and cross-validated using the CINC2017 dataset, with external validation on the CPSC2018 and Chapman datasets. The training process consisted of 100 epochs with a batch size of 256. We employed the AdamW optimizer [37] with a learning rate of 0.001 and a weight of decay of 0.01.

Pre-training, Fine-tuning, and Validation. CTRhythm was initially pre-trained on the MIMIC-IV-ECG dataset. For the pre-training step, we used the MOCO v2 framework [38] with 20 epochs and a batch size of 256. The AdamW optimizer was used with a learning rate of 0.001 and a weight decay of 0.01. For the model fine-tuning on CINC2017, the learning rate was adjusted to 0.0001, while other settings remained the same as in the previous experiment.

V. RESULTS AND DISCUSSION

A. AF detection on CINC2017

We compared CTRhythm with the most current ECG classification methods applied to the CINC2017 dataset. Table I summarizes the F1 scores for all compared methods. CTRhythm achieved a state-of-the-art overall F1 score of 0.831, outperforming all other methods.

B. External Validations

We used two external validation datasets to assess and compare the generalizability of these models. Table II presents the results. Notably, CTRhythm outperformed all other models, achieving the highest accuracy, 0.899 in CPSC2018 and 0.840 in Chapman. Moreover, CTRhythm, trained on data from wearable devices, performed surprisingly well on hospital-collected data, demonstrating its robustness under distribution shifts.

C. Effects of Pre-training

Table III presents the comparative results of CTRhythm with and without pre-training and fine-tuning steps. Notably,

TABLE I. Performance benchmark with the CINC2017 dataset

Method	Model	F1 _N	F1 _A	F1 _O	F1 _{All}
Datta et al. [32]	AdaBoost	0.910	0.798	0.772	0.826
Zihlmann et al. [19]	CNN	0.878	0.790	0.701	0.790
	CRNN	0.888	0.764	0.726	0.792
Lee et al. [10]	CNN	0.897	0.811	0.745	0.818
Yu et al. [33]	CNN	0.914	0.776	0.762	0.817
ATI-CNN [39]	CNN	0.883	0.717	0.702	0.767
Xresnet1d101 [40]	CNN	0.881	0.715	0.674	0.757
MSDNN [41]	CNN	0.900	0.779	0.746	0.808
CTRhythm (Ours)	CNN+Transformer	0.910	0.811	0.772	0.831

Footnote: the best results are **bolded**, with **gray** indicating the second best.

TABLE II. Performance benchmark with two external validation datasets

Dataset	Method	F1 _N	F1 _A	F1 _{All}	Accuracy
CPSC2018	ATI-CNN [39]	0.901	0.790	0.846	0.759
	Xresnet1d101 [40]	0.899	0.764	0.831	0.749
	MSDNN [41]	0.887	0.791	0.839	0.775
	CTRhythm	0.968	0.918	0.943	0.899
Chapman	ATI-CNN [39]	0.853	0.797	0.825	0.728
	Xresnet1d101 [40]	0.884	0.801	0.842	0.753
	MSDNN [41]	0.892	0.807	0.849	0.797
	CTRhythm	0.923	0.890	0.906	0.840

Footnote: the best results are **bolded**, with **gray** indicating the second best.

by incorporating ECG big data for pre-training, CTRhythm's performance was enhanced by 1% in accuracy.

TABLE III. Effects of model pre-training with MIMIC-IV-ECG

	F1 _N	F1 _A	F1 _O	F1 _{All}	Accuracy
Random Init.	0.910	0.811	0.772	0.831	0.854
Pre-trained	0.916	0.817	0.788	0.840	0.862

D. Visualization of Learned Features

To gain insights into the learned latent ECG features, we visualized the attention values from intermediary layers using the Attention Rollout technique [42]. The visualized attention mechanisms showed that CTRhythm learned interpretable patterns. For instance, we displayed the first samples of AF (Rec. A00004) and Normal (Rec. A00001) (see Fig. 3).

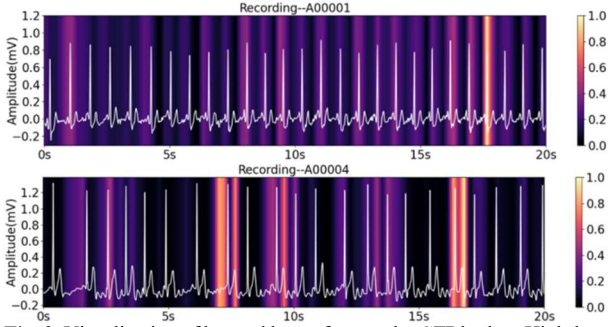


Fig. 3. Visualization of learned latent features by CTRhythm. High-heat areas are the time points where CTRhythm gives high attention.

As seen in the Normal ECG (Fig. 3 upper panel), there are discernible periodic P waves that indicate regular atrial depolarization, and CTRhythm attends regularly to those spans in the QRS complexes. In the AF ECG (Fig. 3 lower panel), the atrial activation is chaotic, discernible P waves are absent, and RR intervals are notably irregular. As a result, the attention heat was enriched at such time points irregularly.

E. Ablation Studies

Ablation of Model Architecture. We performed ablation studies on CTRhythm's *CNN+Transformer* architecture to assess the effect of using Transformer. The *CNN* model was

made by removing the Transformer module from CTRhythm. The *CRNN* model was made by integrating the *CNN* model with two LSTM layers. We created a *CNN-Larger* model by augmenting the convolutional kernels in the *CNN* model but keeping its parameters in scale with CTRhythm. The cross-validated performance of all these models is presented in Table IV. The *CNN+Transformer* encoder module stands out in showing the highest accuracy, in particular achieving the highest F1 score in detecting AF.

TABLE IV. Model architecture ablation

Architecture	F1 _N	F1 _A	F1 _O	F1 _{All}	Accuracy
CNN	0.895	0.793	0.709	0.799	0.830
CNN+LSTM	0.912	0.756	0.756	0.808	0.844
CNN-Larger	0.902	0.782	0.738	0.807	0.839
CNN+Transformer	0.910	0.811	0.772	0.831	0.854

Ablation of Model Size. We also conducted an ablation study on model size, focusing on the number of Transformer encoder layers, and the result was shown in Table V. CTRhythm with 6 Transformer layers achieved the highest overall F1 score (0.831) and accuracy (0.854). Notably, adding two Transformer layers already improved the overall F1 score of the *CNN-only* model by 2% (0.799 vs 0.825). This indicated that even the simplest integration of Transformer to CNN can enhance a model's ability to encode global rhythmic features, thus improving AF detection. Therefore, deploying such a 2-layer CTRhythm to a small wearable device with significantly constrained computational resources is viable.

TABLE V. Model size ablation

Transformer Layers	F1 _N	F1 _A	F1 _O	F1 _{All}	Accuracy
0 (CNN)	0.895	0.793	0.709	0.799	0.830
2	0.909	0.795	0.771	0.825	0.852
4	0.909	0.792	0.765	0.822	0.849
6	0.910	0.811	0.772	0.831	0.854
8	0.908	0.788	0.761	0.819	0.847

F. Computation Requirement Analysis

We designed CTRhythm, anticipating it to be integrated into smartphones or wearable devices. When processing a 60-

second ECG, CTRhythm requires 90 million floating-point operations and occupies ~23 MB of memory. Morden processor, such as the Apple A16 processor has 6 GB of memory and can perform up to 1.79 trillion floating-point operations per second. This implies that smartphones like the iPhone can perform AF detection using CTRhythm within microseconds. For wearable devices that have computation capabilities exceeding a hundred million floating-point operations per second, CTRhythm is directly deployable to such devices.

VI. CONCLUSION

In this study, we presented a novel AF detection model, CTRhythm, which integrates Transformer and CNN encoders and is optimized for single-lead ECG. Unlike previous CNN or CRNN models, CTRhythm incorporated Transformer layers to overcome CNN's limits due to localized reception. By effectively encoding local and global patterns, CTRhythm outperformed all competing models in cross-validation and showed the highest generalizability in independent external validations. Additionally, pre-training on a large hospital-based ECG dataset enhanced CTRhythm's performance on single-lead data. Our findings highlighted the significance of fusing local and global information for improved heart rhythm classification. The source code for CTRhythm is freely available at <https://github.com/labxscut/CTRhythm>.

ACKNOWLEDGMENTS

This study was funded by the Guangdong Basic and Applied Basic Research Foundation (2022A1515-011426 and 2024A1515-010699) and in part by the National Natural Science Foundation of China (61873027, 81970200, and 82271609), the Guangzhou Municipal Science and Technology Project (2023B01J1011), and the Shenzhen Science and Technology Program (JCYJ20190808100817047 and RCBS20200714114909234).

REFERENCES

- [1] P. Ganesan *et al.*, "Quantifying a spectrum of clinical response in atrial tachyarrhythmias using spatiotemporal synchronization of electrograms," *Europace*, vol. 25, no. 5, p. euad055, 2023.
- [2] R. B. Schnabel *et al.*, "Early diagnosis and better rhythm management to improve outcomes in patients with atrial fibrillation: the 8th AFNET/EHRA consensus conference," *Europace*, vol. 25, no. 1, pp. 6-27, 2023.
- [3] K. Bayoumy *et al.*, "Smart wearable devices in cardiovascular care: where we are and how to move forward," *Nature Reviews Cardiology*, vol. 18, no. 8, pp. 581-599, 2021.
- [4] K. C. Siontis, P. A. Noseworthy, Z. I. Attia, and P. A. Friedman, "Artificial intelligence-enhanced electrocardiography in cardiovascular disease management," *Nature Reviews Cardiology*, vol. 18, no. 7, pp. 465-478, 2021.
- [5] H. Serhal, N. Abdallah, J.-M. Marion, P. Chauvet, M. Oueidat, and A. Humeau-Heurtier, "Overview on prediction, detection, and classification of atrial fibrillation using wavelets and AI on ECG," *Computers in Biology and Medicine*, vol. 142, p. 105168, 2022.
- [6] O. M. A. Ali, S. W. Kareem, and A. S. Mohammed, "Evaluation of electrocardiogram signals classification using CNN, SVM, and LSTM algorithm: A review," in *2022 8th International Engineering Conference on Sustainable Technology and Development (IEC)*, 2022: IEEE, 2022, pp. 185-191.
- [7] Y. Xia, N. Wulan, K. Wang, and H. Zhang, "Detecting atrial fibrillation by deep convolutional neural networks," *Computers in Biology and Medicine*, vol. 93, pp. 84-92, 2018.
- [8] A. Y. Hannun *et al.*, "Cardiologist-level arrhythmia detection and classification in ambulatory electrocardiograms using a deep neural network," *Nature Medicine*, vol. 25, no. 1, pp. 65-69, Jan 2019.
- [9] Z. Xiong, M. K. Stiles, and J. Zhao, "Robust ECG signal classification for detection of atrial fibrillation using a novel neural network," in *2017 Computing in Cardiology (CinC)*, 2017: IEEE, pp. 1-4.
- [10] H. Lee and M. Shin, "Learning explainable time-morphology patterns for automatic arrhythmia classification from short single-lead ECGs," *Sensors*, vol. 21, no. 13, p. 4331, 2021.
- [11] B. S. Chandra, C. S. Sastry, S. Jana, and S. Patidar, "Atrial fibrillation detection using convolutional neural networks," in *2017 Computing in Cardiology (CinC)*, 2017: IEEE, pp. 1-4.
- [12] J. Rubin, S. Parvaneh, A. Rahman, B. Conroy, and S. Babaeizadeh, "Densely connected convolutional networks for detection of atrial fibrillation from short single-lead ECG recordings," *Journal of Electrocardiology*, vol. 51, no. 6, pp. S18-S21, 2018.
- [13] F. Andreotti, O. Carr, M. A. Pimentel, A. Mahdi, and M. De Vos, "Comparing feature-based classifiers and convolutional neural networks to detect arrhythmia from short segments of ECG," in *2017 Computing in Cardiology (CinC)*, 2017: IEEE, pp. 1-4.
- [14] A. Vaswani *et al.*, "Attention is all you need," *Advances in Neural Information Processing Systems*, vol. 30, 2017.
- [15] C. Chen, Z. Hua, R. Zhang, G. Liu, and W. Wen, "Automated arrhythmia classification based on a combination network of CNN and LSTM," *Biomedical Signal Processing and Control*, vol. 57, 2020.
- [16] J. Rahul and L. D. Sharma, "Automatic cardiac arrhythmia classification based on hybrid 1-D CNN and Bi-LSTM model," *Biocybernetics and Biomedical Engineering*, vol. 42, no. 1, pp. 312-324, 2022.
- [17] M. Limam and F. Precioso, "Atrial fibrillation detection and ECG classification based on convolutional recurrent neural network," in *2017 Computing in Cardiology (CinC)*, 2017: IEEE, pp. 1-4.
- [18] J. Zhang, A. Liu, M. Gao, X. Chen, X. Zhang, and X. Chen, "ECG-based multi-class arrhythmia

- detection using spatio-temporal attention-based convolutional recurrent neural network," *Artificial Intelligence in Medicine*, vol. 106, p. 101856, Jun 2020.
- [19] M. Zihlmann, D. Perekrestenko, and M. Tschannen, "Convolutional recurrent neural networks for electrocardiogram classification," in *2017 Computing in Cardiology (CinC)*, 2017: IEEE, pp. 1-4.
- [20] J. D. M.-W. C. Kenton and L. K. Toutanova, "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding," in *Proceedings of NAACL-HLT*, 2019, pp. 4171-4186.
- [21] A. Dosovitskiy *et al.*, "An image is worth 16x16 words: Transformers for image recognition at scale," *arXiv preprint arXiv:2010.11929*, 2020.
- [22] D. Yun *et al.*, "Automatic segmentation of atrial fibrillation and flutter in single-lead electrocardiograms by self-supervised learning and Transformer architecture," *Journal of the American Medical Informatics Association*, 2023.
- [23] M.-U. Yang, D.-I. Lee, and S. Park, "Automated diagnosis of atrial fibrillation using ECG component-aware transformer," *Computers in Biology and Medicine*, vol. 150, p. 106115, 2022.
- [24] R. Hu, J. Chen, and L. Zhou, "A transformer-based deep neural network for arrhythmia detection using continuous ECG signals," *Computers in Biology and Medicine*, vol. 144, p. 105325, 2022.
- [25] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Computation*, vol. 9, no. 8, pp. 1735-1780, 1997.
- [26] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2016, pp. 770-778.
- [27] S. Ioffe and C. Szegedy, "Batch normalization: Accelerating deep network training by reducing internal covariate shift," in *International Conference on Machine Learning*, 2015: pmlr, pp. 448-456.
- [28] V. Nair and G. E. Hinton, "Rectified linear units improve restricted boltzmann machines," in *Proceedings of the 27th International Conference on Machine Learning (ICML-10)*, 2010, pp. 807-814.
- [29] J. L. Ba, J. R. Kiros, and G. E. Hinton, "Layer normalization," *arXiv preprint arXiv:1607.06450*, 2016.
- [30] M. Lin, Q. Chen, and S. Yan, "Network in network," *arXiv preprint arXiv:1312.4400*, 2013.
- [31] G. D. Clifford *et al.*, "AF classification from a short single lead ECG recording: The PhysioNet/computing in cardiology challenge 2017," in *2017 Computing in Cardiology (CinC)*, 2017: IEEE, pp. 1-4.
- [32] S. Datta *et al.*, "Identifying normal, AF and other abnormal ECG rhythms using a cascaded binary classifier," in *2017 Computing in Cardiology (CinC)*, 2017: IEEE, pp. 1-4.
- [33] Z. Yu *et al.*, "DDCNN: A Deep Learning Model for AF Detection From a Single-Lead Short ECG Signal," *IEEE Journal of Biomedical and Health Informatics*, vol. 26, no. 10, pp. 4987-4995, 2022.
- [34] F. Liu *et al.*, "An open access database for evaluating the algorithms of electrocardiogram rhythm and morphology abnormality detection," *Journal of Medical Imaging and Health Informatics*, vol. 8, no. 7, pp. 1368-1373, 2018.
- [35] J. Zheng *et al.*, "Optimal multi-stage arrhythmia classification approach," *Scientific reports*, vol. 10, no. 1, p. 2898, 2020.
- [36] A. E. Johnson *et al.*, "MIMIC-IV, a freely accessible electronic health record dataset," *Scientific data*, vol. 10, no. 1, p. 1, 2023.
- [37] I. Loshchilov and F. Hutter, "Decoupled Weight Decay Regularization," in *International Conference on Learning Representations (ICLR)*, 2019.
- [38] K. He, H. Fan, Y. Wu, S. Xie, and R. Girshick, "Momentum contrast for unsupervised visual representation learning," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2020, pp. 9729-9738.
- [39] Q. Yao, R. Wang, X. Fan, J. Liu, and Y. Li, "Multi-class arrhythmia detection from 12-lead varied-length ECG using attention-based time-incremental convolutional neural network," *Information Fusion*, vol. 53, pp. 174-182, 2020.
- [40] N. Strodthoff, P. Wagner, T. Schaeffter, and W. Samek, "Deep learning for ECG analysis: Benchmarks and insights from PTB-XL," *IEEE journal of biomedical and health informatics*, vol. 25, no. 5, pp. 1519-1528, 2020.
- [41] J. Lai *et al.*, "Practical intelligent diagnostic algorithm for wearable 12-lead ECG via self-supervised learning on large-scale dataset," *Nature Communications*, vol. 14, no. 1, p. 3741, 2023.
- [42] S. Abnar and W. Zuidema, "Quantifying attention flow in transformers," *arXiv preprint arXiv:2005.00928*, 2020.